

REPORT

Title

Implementation of the Crypt Arithmetic Problem Using Python

Aim

To solve the Crypt Arithmetic puzzle by assigning unique digits to letters such that the arithmetic equation is satisfied.

Problem Statement

A Crypt Arithmetic problem is a mathematical puzzle where digits are replaced by letters. The objective is to determine the digit corresponding to each letter while ensuring that every letter has a unique digit, no leading digit is zero, and the arithmetic operation is valid. The puzzle **SEND + MORE = MONEY** is one of the most popular examples of a Crypt Arithmetic problem.

Requirements

- Python 3.x

Procedure

1. Define the letters used in the puzzle.
2. Generate all possible digit assignments.
3. Ensure each letter has a unique digit.
4. Prevent leading letters from being assigned zero.

5. Compute the numerical values of SEND, MORE, and MONEY.
6. Check whether the equation **SEND + MORE = MONEY** is satisfied.
7. Display the valid solution.

Python Program

```
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')

for p in permutations(range(10), len(letters)):
    S, E, N, D, M, O, R, Y = p

    if S == 0 or M == 0:
        continue

    send = 1000*S + 100*E + 10*N + D
    more = 1000*M + 100*O + 10*R + E
    money = 10000*M + 1000*O + 100*N + 10*E + Y

    if send + more == money:
        print("Solution Found")
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Sample Output

Solution Found

SEND = 9567

MORE = 1085

MONEY = 10652

Result

The Crypt Arithmetic puzzle **SEND + MORE = MONEY** was solved successfully by assigning unique digits to each letter, satisfying all puzzle constraints.

Conclusion

The Crypt Arithmetic problem demonstrates the application of logical reasoning and constraint satisfaction techniques. It is widely used in Artificial Intelligence to illustrate search strategies and problem-solving methods.